

CASE A



UNIVERSITY OF AMSTERDAM
Economics & Business



VSAE



**ECONOMETRIC
GAME**

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Econometric Game 2023: Predict the Hotspots of Child Malnutrition

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1 Introduction

Wasting, an acute form of malnutrition increases a child's risk of infection and death and reduces their ability to learn and be productive [1, 2, 3]. Children aged between 6-59 months form the cohort for studying wasting [1, 2, 3, 4]. As per UNICEF and WHO, when the ratio of weight to the height of children less than 5 years is less than -3 Z-score from the median of the World Health Organization's (WHO's) Child Growth Standards, it is defined as a case of severe wasting. The latest United Nations (UN) report estimates 45.4M children under 5 years were wasted globally, and over 13.6M were severely wasted in 2020.

Prevention of malnutrition needs effective and coordinated action adapted to the needs of the population in terms of human security, access to healthcare, food security and nutrition. Estimates of caseloads of wasting are necessary for prioritizing the most exposed areas and populations at risk of under-nutrition. The burden of wasting for a given time period is driven by the prevalence at the start of the time period and the incidence of new cases in that period [5]. Typically, obtaining incidence estimates is much more expensive compared to obtaining prevalence estimates, which are more readily available to program managers.

In the absence of incidence data, the burden can be estimated as a function of only prevalent cases and the Incidence Correction Factor (ICF)¹. This method to estimate the burden of severe wasting in the absence of incidence data has formed the basis for UN guidance and other published work to estimate caseloads for severe wasting treatment programs [1, 5, 6]. To improve program planning, global guidance recommends a single incidence correction factor of 1.6 be applied to available prevalence estimates to account for incident cases. However, [1] observe that the incidence correction factors vary between sites and that the burden of severe wasting will often be underestimated using

1

Prevalence = *Incidence* $\times$ *d*, where *d* = Average duration of untreated illness

Incidence = *Prevalence* $\times$ $(\frac{t}{d})$ = *Prevalence* $\times$ *ICF*, where *t* = period for which incidence is estimated

Burden = *Prevalent Cases* + *Incident Cases* = *Population*_{6-59months} $\times$ *Prevalence*(1 + *ICF*);

the currently recommended incidence correction factor of 1.6. The variation in the true incidence correction factor across various sites and contexts results in inaccurate estimates of the caseload. Hence, there is a need for improved burden estimates of wasting in order to better prepare interventions for the prevention and treatment of acute malnutrition.

2 Your task

The task is to predict the **burden** ($Burden = Prevalent\ Cases + Incident\ Cases$) of wasting or level of malnutrition. The target variable (*target* in the dataset) is the level of malnutrition (based on **Burden**, especially *SAM Burden*) in each administrative region of the Sahel (G5+1 countries). The task is a multiclass classification where the target (in Table 2) takes one of ‘**Low**’, ‘**Medium**’, ‘**High**’, or ‘**Very High**’ values. These classes have been mapped to (0,1,2,3), i.e. ‘**Low**’ $\rightarrow$ 0, ‘**Medium**’ $\rightarrow$ 1, ‘**High**’ $\rightarrow$ 2, and ‘**Very High**’ $\rightarrow$ 3. The features come from nutritional surveys from the Sahel region and from a publicly available INFORM dataset². The detailed description of the features are presented in Appendix A in Table 2 and Table 1.

It is your job to come up with a predictive algorithm that can predict levels of malnutrition in the Sahel region of Africa which can help in early identification and hence timed interventions based on needs of the population. It is entirely up to you how to make use of the available data. You must produce a brief report including:

1. a clear and concise statement of your substantive findings;
2. a discussion and motivation of the steps towards your final specification;
3. an assessment of your predictive model with specifics about choice of model, evaluation metric, validation strategy and identification of factors affecting child malnutrition;
4. a concluding section in which you discuss the implications of your findings.

When submitting your report, send your file to juryeg@vsae.nl. Title your email and name your file the following: Case_A [your university] [your submission number]. Also make sure you don’t mention your university or names anywhere in the report itself. This is important to ensure that the jury will remain unbiased. The deadline for handing in the report is Thursday 20th at 18:00.

References

- [1] Sheila Isanaka, Christopher T Andersen, Simon Cousens, Mark Myatt, André Briend, Julia Krasevec, Chika Hayashi, Amy Mayberry, Louise Mwirigi, and Saul Guerrero. Improving estimates of the burden of severe wasting:

²INFORM

- analysis of secondary prevalence and incidence data from 352 sites. *BMJ global health*, 6(3):e004342, 2021.
- [2] Peter Katona and Judit Katona-Apte. The interaction between nutrition and infection. *Clinical Infectious Diseases*, 46(10):1582–1588, 2008.
 - [3] Christine M McDonald, Karim P Manji, Roland Kupka, David C Bellinger, Donna Spiegelman, Rodrick Kisenge, Gernard Msamanga, Wafaie W Fawzi, and Christopher P Duggan. Stunting and wasting are associated with poorer psychomotor and mental development in hiv-exposed tanzanian infants. *The Journal of nutrition*, 143(2):204–214, 2013.
 - [4] Cesar G Victora, Linda Adair, Caroline Fall, Pedro C Hallal, Reynaldo Martorell, Linda Richter, Harshpal Singh Sachdev, Maternal, Child Undernutrition Study Group, et al. Maternal and child undernutrition: consequences for adult health and human capital. *The lancet*, 371(9609):340–357, 2008.
 - [5] Assaye Bulti, André Briend, Nancy M Dale, Arjan De Wagt, Faraja Chiwile, Stanley Chitekwe, Chris Isokpunwu, and Mark Myatt. Improving estimates of the burden of severe acute malnutrition and predictions of caseload for programs treating severe acute malnutrition: experiences from nigeria. *Archives of Public Health*, 75(1):1–8, 2017.
 - [6] Nancy M Dale, Mark Myatt, Claudine Prudhon, and André Briend. Using cross-sectional surveys to estimate the number of severely malnourished children needing to be enrolled in specific treatment programmes. *Public health nutrition*, 20(8):1362–1366, 2017.

Appendices

A Data Description - Main Project

Table 1: Source and description for deterioration data

Features	Descriptions	Sources
Diarrhea	Passage of three or more loose or watery stools per 24 hr for children U5	SMART Survey
Malaria fever	Death rate associated with the number of deaths caused by malaria per 100,000 people/year	SMART Survey
VITA	Vitamin A Coverage	SMART Survey
Deworming	Deworming among children U5	SMART Survey
Measles	Measles among children U5	SMART Survey
ARI Cough	Cough caused by acute respiratory infections between children U5	SMART Survey
Access to health structures	Access to health structures	Cluster Nutrition data consultation
Access to drinking water no access	Access to drinking water no access	Cluster Nutrition data consultation
CH-Ph 3-5	Cadre Harmonisé total phase 3 to 5 population	Food Insecurity Hotspot Analysis
IDPs	Internally displaced persons, including persons of concern	
dietary diversity (not achieved)	dietary diversity (not achieved)	Cluster Nutrition data consultation
Meal frequency/number of meals (not achieved)	Meal frequency/number of meals (not achieved)	Cluster Nutrition data consultation

Table 2: Source and description for demographic and nutrition data

Features	Descriptions	Sources
Country	Countries in Sahel Regions	Provided by countries
Admin 1	First level of administrative	Provided by countries
ADMIN 2 Administrative	Second level of administrative	Provided by countries
Total population	Total population	Provided by countries
Population 6 to 59 months	Population of children between 6 to 59 months	Provided by countries
SAM incidence correction factor	Severe Acute Malnutrition incidence correction factor	Provided by countries
MAM incidence correction factor	Moderate Acute Malnutrition incidence correction factor	Provided by countries
GAM prevalence	Global Acute Malnutrition incidence correction factor	Nutrition survey
SAM prevalence	Severe Acute Malnutrition prevalence	Nutrition survey
MAM prevalence	Moderate Acute Malnutrition prevalence	Nutrition survey
MAM 6-59 months Prevalence	Moderate Acute Malnutrition for children between 6 to 59 Prevalence	Provided by countries
GAM rate infants U5	Global Acute Malnutrition rate for children under five	SMART Survey
Stunting rate U5	Stunting rate for children under five	SMART Survey
SAM U5 rate	Severe Acute Malnutrition rate for children under five	SMART Survey
GAM Rate FEFA	Global Acute Malnutrition rate for pregnant and lactating women	SMART Survey
GAM BURDEN 2020R	Revised value of GAM burden after Covid-19 pandemic	Nutrition survey
SAM BURDEN 2020R	Revised value of SAM burden after Covid-19 pandemic	Nutrition survey
MAM BURDEN 2020R	Revised value of MAM burden after Covid-19 pandemic	Nutrition survey
GAM BURDEN	Global Acute Malnutrition burden	Nutrition survey
SAM BURDEN	Severe Acute Malnutrition burden	Nutrition survey
MAM BURDEN	Moderate Acute Malnutrition burden	Nutrition survey